

Lab-3: Simulation of Single Queue Server System (M/M/1 Model)

Lab Date: January 21st, 2026

Deadline: January 25th, 2026

Class: CSIT 5th Semester, New Summit College

Subject: Simulation & Modeling

Submission Mode: **Handwritten Document** (Print Output Section Only)

Objective:

To simulate the Single Queue Server System (M/M/1 Mode) and provide a hands-on experience in modeling, analyzing, and optimizing real-world queuing problems using simulation techniques.

Learning Outcomes

After completing this lab, students will be able to:

- To simulate a single-server queue system with Poisson arrivals and exponential service times.
- To measure different performance metrics in M/M/1 Queuing Systems.
- To analyze the impact of arrival and service rates.

Key performance metrics:

Metric	Formula (Theory)	Simulation Meaning
ρ	λ / μ	Fraction of time the server is busy

L	$\lambda / (\mu - \lambda)$	Avg. customers in system
Lq	$\lambda^2 / [\mu(\mu - \lambda)]$	Avg. customers waiting
W	$1 / (\mu - \lambda)$	Avg. time in system
Wq	$\lambda / [\mu(\mu - \lambda)]$	Avg. waiting time
P₀	$1 - \rho$	Probability that the system is empty
Stability	$\rho < 1$	Required for steady state

Tasks:

1. Define the arrival rate (λ) and service rate (μ).
2. Simulate customer arrivals and service times.
3. Track queue length, waiting time, and idle time.
4. Compute server utilization and analyze performance metrics.

Report Structure:

- **Objective:** Clearly state the aim of the lab.
- **Theory:** Include a brief explanation of:
 - a. Poisson arrival process (rate λ).
 - b. Exponential service time (rate μ).

- c. Single Queue Server System (M/M/1 Model)
- d. Performance Metrics in M/M/1 Queuing Model.
- **Procedure:** Outline the steps to simulate the system
- **Code / Implementation**
- **Output**
 - a. **Output / Result Analysis**
- **Conclusion:**

References:

1. https://colab.research.google.com/drive/1MKR6285G4vk8AcXh0Sb_MFG25Yu5jwmr?usp=sharing
2. https://drive.google.com/file/d/1e_25hRdyuxb61XxMIhJ9pukjhKka4Re7/view?usp=sharing